Number Of Ways To Traverse Graph

AlgoExpert · Medium · Dynamic Programming

The problem

- Paths from top-left to bottom-right of a width x height grid moving only right/down.
- Equals C(w+h-2, w-1), computed incrementally to avoid overflow.

Intuition

- Paths to a cell come from above or from the left; this is also a binomial coefficient.

Approach

- 1. First row and column have one way
- 2. Each cell sums its top and left neighbours
- 3. Return the bottom-right cell

Java solution

```java
public class NumberOfWaysToTraverseGraph {
    public static long numberOfWaysToTraverseGraph(int width, int height) {
        int n = width + height - 2, k = Math.min(width, height) - 1;
        long r = 1;
        for (int i = 1; i <= k; i++) r = r * (n - k + i) / i;
        return r;
    }
}
```

Complexity

- Time: O(min(w, h))
- Space: O(1)

Recap

- Paths to a cell come from above or from the left; this is also a binomial coefficient.
- Complexity: O(min(w, h))